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Study of Discrete PID Controller for DC Motor Speed Control Using MATLAB

S. Balamurugan,
Assistant Professor (SG),
Department of Electronics and Instrumentation Engineering,
K.L.N College of Engineering,
Tamil Nadu,
Sbalan18@yahoo.com

A. Umarani,
Professor,
Department of Electronics and Instrumentation Engineering,
K.L.N College of Engineering,
Tamil Nadu,
duraiumarani@gmail.com

Abstract- DC motors are known for simple design and consistency are used for control area section of the industry. In many variable speed systems, it is commonly used where wide speed ranges are required. The conventional controller is not compatible with computer controller. To overcome the short falling faced with conventional controller, a simple discrete PID controller designed for the speed control of DC motor. Advantage of discrete controller simple in its design and easily interface with computer system to access the control action. The dynamics of the system is calculated and controller parameters are obtained by root locus method. This research paper emphasizes the design of a simple digital control system that combines a discrete Proportional Integral Derivative (PID) controller with Direct current (DC) motor. The complete simulation is carried out with MATLAB / SIMULINK software.

Key Words- DC motor speed control, discrete PID controller, variable speed system, industrial application

1. INTRODUCTION

DC Motor transforms electricity energy to mechanical energy and it act as a power actuator. It has a rotation armature winding, non-rotating armature magnetic field, permanent magnet which generates different magnetic field and armature connections. Winding develops a different intrinsic speed and helps in regulating the torque. DC motor speed is controlled either by changing armature current or changing variable resistance in armature circuit or field circuit. The above mentioned DC motor come under traditional speed control method.

To increase the speed of traditional DC motor, it can be upgraded by integrating the DC motor with power electronics circuits. Thereafter the performance of the DC motor speed control system gets improved and tracks the desired speed. Therefore the proposed DC motor plays a vital role in the industrial applications where adjustable speed control action is required such as electric cranes robotic and manipulative vehicles.

Metin Demirtas analyzed a Proportional Integral (PI) speed controller for a PLDC motor on off-line and the response of the controller has been reviewed. K. Premkumar *et. al* designed a soft computing technology with PID controller for DC Motor system and the results were analyzed in time domain with varying speed and load conditions. Saqib *et. al* examined antiwindup controller with DC Motor for speed control and the performance of the motor has been checked for closed loop stability. Dayarnab Baidya *et. al* has adopted Fuzzy based Model Reference Adaptive Control (MRAC) for DC motor speed control and the implementation were done with first-order system with second-order system by MIT Rule. Adel A.El-samahy *et. al* implemented Fuzzy based PID with MRAC for DC Motor Speed control and compared Conventional PID against proposed control technique [1 – 10].

The conventional PID controller is of analog system which need additional module to interface with digital computers. The proposed idea of Discrete PID Controller of DC Motor Speed Control is to easily interface the digital computers with the motors. Root – Locus method has been adopted in implementation of Discrete PID controller. This paper includes section 1 briefing the existing DC Motor different speed motor control technology. Section 2 includes DC Motor Mathematical Modeling, Section 3 explaining the conventional PID controller design and discrete PID controller design, Section 4 comprises of results and discussion of different controllers and Section 5 on conclusion [1 – 23].

2. Mathematical model of simple DC mo

Figure 1 shows DC Motor equivalent circuit with armature resistance (R_a), self inductance (L_a), induced emf (e).

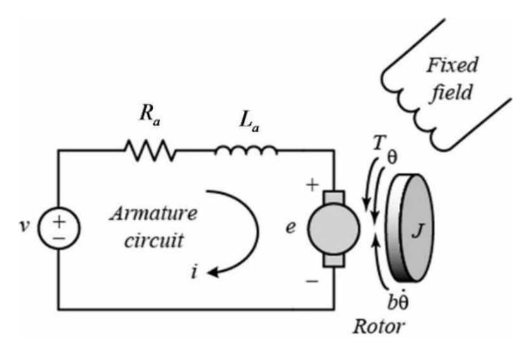

Figure 1 Equivalent circuit of DC motor

The terminal voltage (V_t) can be determine as

$$V_t = E_b + R_a . I_a + L_a . \frac{dI_a}{dt} \tag{1}$$

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In steady state condition, armature current is zero and rate of armature current change is also zero. Equation (1) becomes

$$V_t = E_b + R_a.I_a \tag{2}$$

Air gap power is represented in electromagnetic speed and torque as

$$P_a = \omega_m T = E_b.I_a \tag{3}$$

Where ω_m is the motor speed
T is the torque
I_a is the armature current

Therefore, electromagnetic torque is written as

$$T = \frac{E_b.I_a}{\omega_m} \tag{4}$$

The back emf (E_b) is given by

$$E_b = K_b \omega_m \tag{5}$$

Therefore,

$$T = K_b I_a \tag{6}$$

Moment of inertia J of DC motor, viscous friction coefficient B and an acceleration torque T_a is provided by:

$$T_a = T - Tl = J.\frac{d\omega_m}{dt} + B.\omega_m \tag{7}$$

Where T_l is load torque. Eq. (1) and (6) describe dynamics of DC Motor with load condition. With Laplace Transform for Eq. (1) & Eq (6), gives

$$I_a(s) = \frac{V(s) - K_b.\omega_m(s)}{R_a + sL_a} \tag{8}$$

$$\omega_m(s) = \frac{K_b I_a(s) - T_l(s)}{B + sJ} \tag{9}$$

The open-loop transfer function of DC motor's speed is expressed as:

$$\frac{\theta}{V} = \frac{K}{(Js+b)+(Ls+R)+K^2} \tag{10}$$

Open loop response of DC motor with step input is as shown in Fig 2. Simulink Model and response of DC motor for input voltage, motor current and motor speed is shown in the Figure 3(a) & 3(b). The response was obtained for the dynamics of the system described in the Equations (8) and (9).

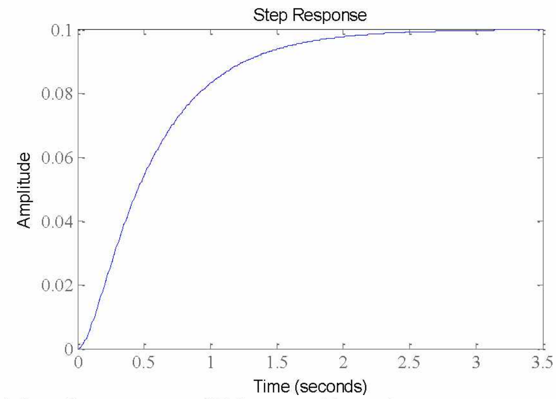
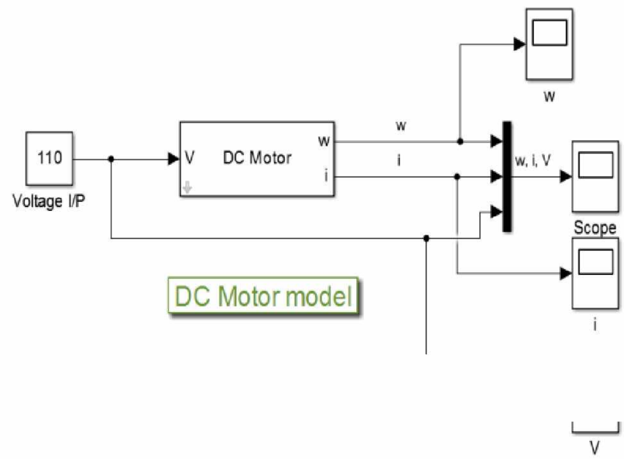
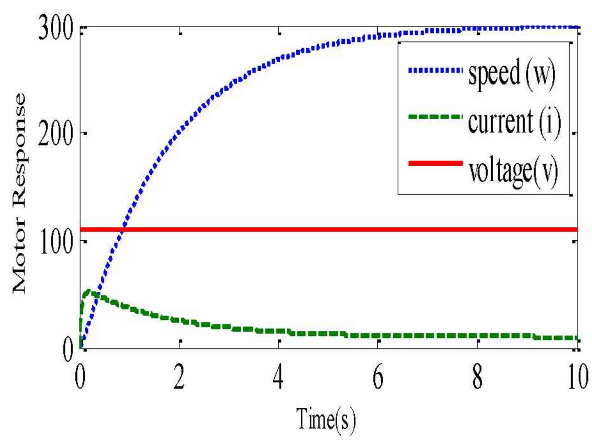


Figure 2 Open loop response of DC motor with step input



(a)



(b)

Fig. 3. (a & b) Simulink Model and Response of DC Motor

The specifications adopted to simulate the DC motor have been enlisted in the below Table1. The tabulated specifications were used to represent the simulation of DC motor.

TABLE 1: SPECIFICATION OF DC MOTOR

Parameters	Values and units
Resistance (R_a)	1 ohm
Inductance (L_a)	0.5 H
Electromotive force constant (K_e=K_t)	0.01 Nm/Amp
Moment of inertia of rotor (J)	0.01 kg*m^2/s^2
Damping ratio of mechanical system (B)	0.1 Nms
Input (V)	Source Voltage
Output (theta dot)	Rotating speed
Rotor and shaft	Rigid

3. Controller Design Procedure

3.1. PID Controller

PID controllers are used in wide range of industrial applications. Around 95 percent of the industry's closed loop operations use PID controllers.PID stands for Proportional- Integral- Derivative. Such three controllers are combined in such a way that a control signal is produced. The general form of an PID controller is given by

$$u(t) = K_p.e(t) + K_i \int_0^t e(t).dt + K_d.\frac{de}{dt} \quad (11)$$

where,

Kp → Proportional Gain,

Ki → integral Gain,

Kd → derivative Gain

PID controller functions with closed-loop system. Variable (e) specifies tracking error, difference among desired output (r) and actual output (y). Error signal (e) is provided to PID controller, and controller evaluates derivative and integral of error signal with time. Control signal (u) is equal to proportional gain (K_p) times magnitude of error plus integral gain (K_i) times integral of error and derivative gain (K_d) times error derivative.

There are several PID controller structures and it depends upon the manufacturers. However only two topologies are used frequently: Parallel (non-interactive) and Series (interactive). PID controller is employed for SISO and MIMO systems. Numerous systems comprise interconnected loops. Controller design is successfully solved by traditional MIMO techniques. MIMO drawback results in state space high order controllers. In addition, the systems contain non negligible time delays which cannot be handled easily. Considering all these drawbacks SISO procedures for decentralized PID controllers tuning for MIMO systems was employed.

Let DC motor is designed for 1 rad /sec step input with the following transient parameters

1 Settling time : < 2 sec
2 Overshoot : < 5%
3 Steady-state error: < 1%

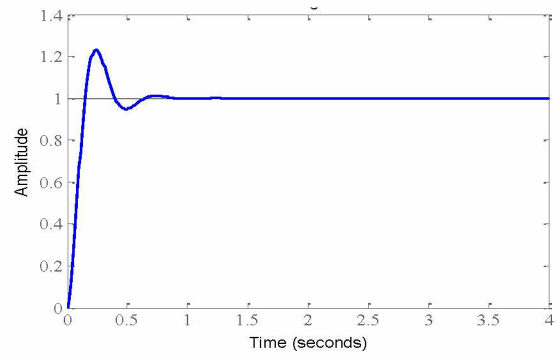


Fig. 4. Continuous PID Control System with DC motor

The closed loop transient response of the system for the above designed parameter is in Fig 4.

Transient performance of speed response indicates large peak overshoot, K_i, and small K_d.

3.2 *Discrete PID controller*

Initial process of modeling discrete control system is to transform continuous transfer function to discrete transfer function with sampling time (T_s) and hold circuit. Here, hold used as zero-order hold ('zoh'). Transfer function for discrete speed control system is expressed as

$$\frac{\theta(z)}{V(z)} = \frac{0.0092z + 0.0057}{z^2 - 1.0877z - 0.2369}$$

Where

$\theta(Z)$ = Motor Angle, V (Z) =Motor input voltage

Open loop response with step input for discrete DC motor is shown in the Figure 5:

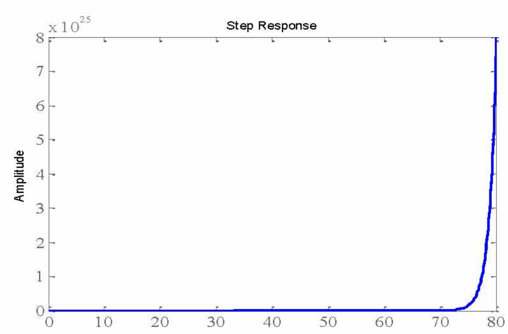


Fig. 5. Open loop response of Discrete DC motor

To obtain closed loop response, consider sampling time $\mathbf{T_s}$ as 0.12 sec, where 1/10 the time constant of system and settling time is 2 sec. Closed loop response of discrete DC motor is depicted as fig 6.

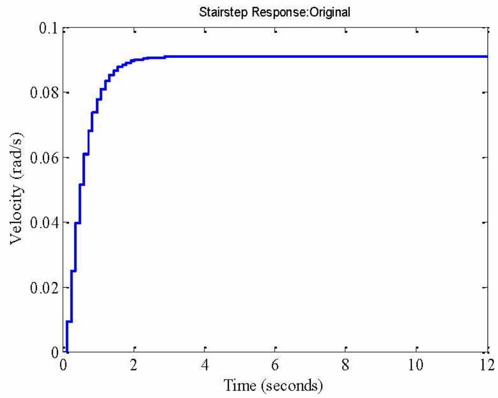


Fig. 6. Closed loop response of Discrete DC motor

Figure 6 depicts closed loop system is unstable. Therefore, system demands for compensator to attain stability of system. Root locus technique is handled to design required compensator. Figure.7 shows the response of discrete PID controller of DC motor

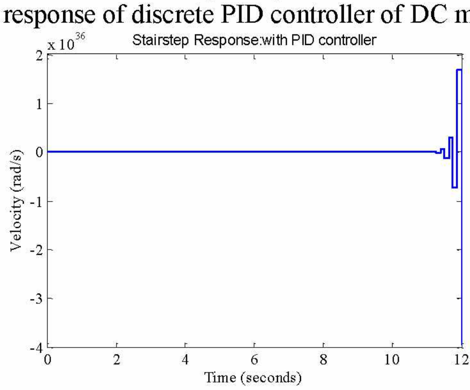


Fig. 7 Discrete PID controller response for DC motor

From Figure 8, it is noticed that the dominant poles has to be moved towards origin and obtain the better discrete PID controller gain. This gain attainment ensures the stability of the system due to the change in the location of dominant pole. The dominant pole is located to cancel the zero at -0.62. Here, an appropriate gain from root locus to fulfill design requirements with compensator.

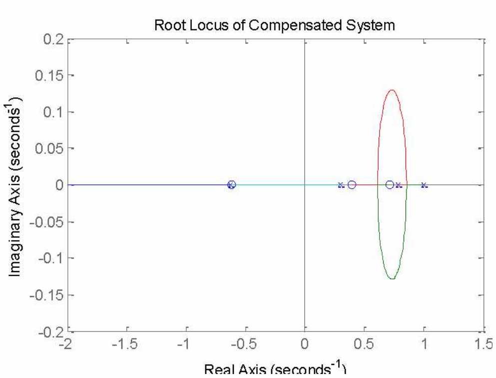


Fig. 8. Root locus plot of DC motor

The appropriate gain obtained through MATLAB can be implemented to the closed loop and system response is shown in Fig 9.

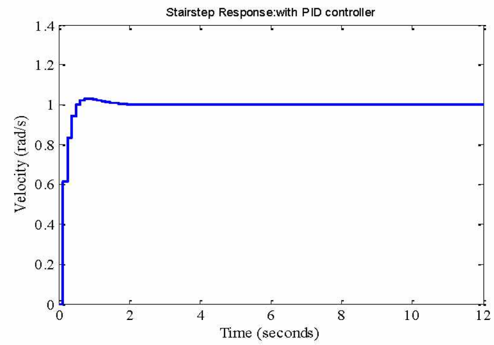


Fig. 9. Discrete PID with DC motor response

4. Results and Discussion

DC Motor with Discrete PID Controller in MATLAB Simulink environment is shown in Figure 10

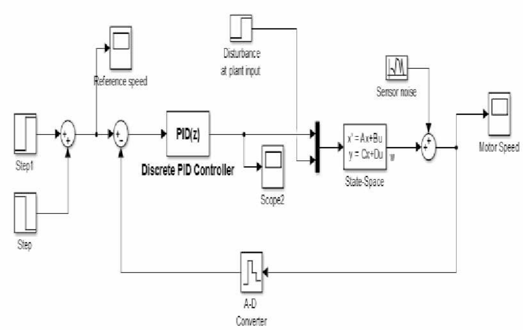
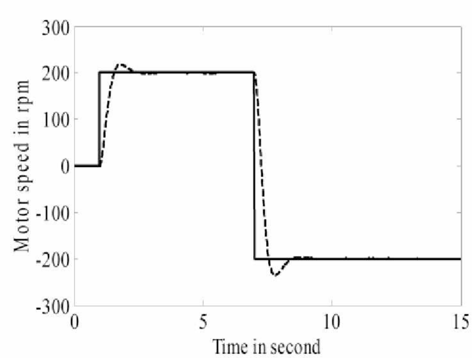
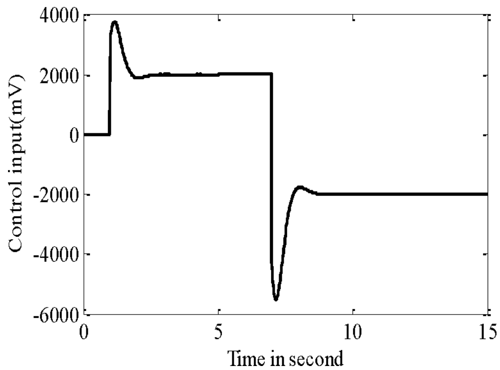


Fig. 10. Simulink representation of discrete PID control system with DC motor

The above Simulink representation is simulated and the obtained results are shown in the Figure 11(a & b)



(a) Motor speed response



(b) Control input response

Fig. 11. Discrete PID Controller response

Figure 11(a) depicts the different desired speed and track the reference speed. Figure 11(b) represent Motor control input for various applied voltage.

The results of the conventional and discrete controllers were analyzed and with the information inferred, they are compared in the time domain. The table of comparison is depicted in Table 2.

TABLE 2 COMPARISON OF CONVENTIONAL AND DISCRETE PID CONTROLLER IN TIME DOMAIN

Time domain parameter	Conventional PID Controller	Discrete PID controller
Rise Time(s)	1.6100	0.3600
Settling Time(s)	8.9500	2.3600
Peak overshoot (%)	16.5567	9.0383
Peak Time(s)	3.6000	1.8000

From the results, Discrete PID controller provides better time domain parameter enhancing performance and system efficiency. It is noted that Discrete PID controller has a reduced rise time of 0.3600 sec against PID 1.6100 sec. And it is found that Discrete PID controller has reduced Peak over shoot of 9.0383% over PID controller with 16.5567%. Hence it is clearly noticed that settling time, rise time and peak time are greatly reduced in the case of Discrete PID controller method compared to conventional PID controller.

5. CONCLUSION

DC motor mathematical model has been represented. Traditional and discrete PID controller has been implemented for a DC motor speed control system. The performance curves attained from conventional and discrete PID controller for DC motor in Simulink environment and the results are compared with time domain specification. It inspects discrete PID controller performance which is better than conventional PID controller. In future, DC motor research may be carried out with hybrid soft computing techniques.

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